HOMEWORK 8.7 · UNIT 8 · NC.8.G.3 · NC.8.G.4

Unit 8 Review

Targets: G.3-2, G.3-3, G.3-4, G.4-6, G.3-6, G.4-7, G.3-7, G.4-8, G.3-9

Name _____ Date _____ Block _____

Remember: Translate $(x + a, y + b)$ · Reflect: x-axis $(x, -y)$, y-axis $(-x, y)$ · Rotate: 90° $(-y, x)$, 180° $(-x, -y)$, 90° clockwise $(y, -x)$ · Dilate (kx, ky). Rigid moves → **congruent** (≅). Add a dilation → **similar** (~). $k = \frac{\text{image}}{\text{pre-image}}$.

SCORE 3.0 Approaching · the basics

Targets: G.3-2, G.3-3, G.3-4, G.4-6

1 Which rule moves a figure 5 units left and 2 units up? REVIEW 8.1 G.3-2

Ⓐ $(x + 5, y - 2)$ Ⓑ $(x - 5, y + 2)$
Ⓒ $(x - 2, y + 5)$ Ⓓ $(x + 5, y + 2)$

2 Reflect $(6, -1)$ over the y-axis. REVIEW 8.1 G.3-3

3 Rotate $(-4, 3)$ 180° about the origin. REVIEW 8.2 G.3-3

Ⓐ $(4, -3)$ Ⓑ $(-4, -3)$ Ⓒ $(4, 3)$ Ⓓ $(-3, -4)$

4 Dilate $(-9, 12)$ by $k = \frac{2}{3}$ about the origin. REVIEW 8.3 G.3-4

5 A figure is rotated 90°, then dilated by $k = 4$. The image is... REVIEW 8.5 G.4-6

congruent | similar, not congruent

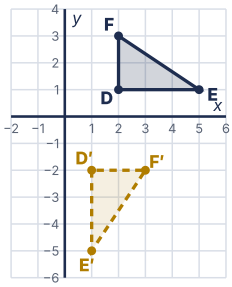
Why? _____

SCORE 4.0 Proficient · on target

Targets: G.3-6, G.4-7, G.3-7

6 Which transformation maps △DEF onto △D′E′F′?

REVIEW 8.2 G.3-6



Ⓐ Rotation 90° clockwise
Ⓑ Rotation 90° counterclockwise
Ⓒ Reflection over the *x*-axis
Ⓓ Rotation 180°

7 A 5 in by 8 in picture is enlarged. The long side becomes 20 in. How long is the short side?

REVIEW 8.6 G.4-7

8 **Pat** made a mistake. Circle it. Explain it. Fix it.

REVIEW 8.4 G.3-7

PAT'S WORK

$A(3, 6) \rightarrow A'(1, 2)$

Scale factor $= 3 \div 1 = 3$

SCORE 5.0 Mastery · go deeper

Targets: G.4-8, G.3-9

9 △ABC has A(1, 1), B(2, 1), C(1, 3). △DEF has D(−2, 2), E(−4, 2), F(−2, 6). Describe a sequence that maps △ABC onto △DEF. Are they congruent or similar?

REVIEW 8.5 G.4-8

First _____ . Then _____ . They are _____ because _____ .

10 A 3 by 5 rectangle is dilated by $k = 2$. Find the new side lengths and the new area. How many times bigger is the area?

REVIEW 8.4 G.3-9

The new sides are _____ . The area is _____ times bigger because _____ .